Reference-Guided Merge Tree Maps: Quantifying Position–Topology Trade-offs

Category: Research

ABSTRACT

Merge tree maps summarize time-varying scalar fields by arranging features along one spatial axis and stacking the resulting profiles over time. Keeping each subtree contiguous preserves merge relations but can displace features from a fixed geographic reference. We introduce Reference-Guided Merge Tree Maps (RG-MTM) to quantify and control this position–topology trade-off. An attained positional certificate separates displacement caused by interval packing from the additional cost of hierarchy. An optimal fixed-order scalar construction retains extremum values and minimizes labelled pairwise merge-level error, defining exact attainable frontiers for small trees. RG-MTM-P uses these certificates to align features while preserving the original merge levels; RG-MTM-R relaxes selected contiguity constraints and reports the resulting scalar distortion. Across ERA5, Fire, OLR, PM2.5, Ring, and Gaussians, comprising 342 frames, hierarchy cost exceeds 2% of the reference extent in 12–63% of real-data frames. Both modes reduce reference-order error relative to TMTM and ST-MTM on every sequence. Relaxation further reduces ERA5 order error from 3.77% to 1.16%, with 1.47% mean and 26.07% maximum normalized merge-level error. ERA5 motion-proxy errors decrease from ST-MTM's 43/104 to 19/104 events, whereas OLR retains the preserving mode's 3/56 errors despite improved alignment. RG-MTM supports explicit selection between merge-level fidelity and reference-position accuracy in offline temporal summaries with small retained feature sets.

Index Terms: Temporal scalar fields, merge tree maps, reference-guided layout, positional certificates, merge-level distortion.

1 INTRODUCTION

1.1 Static Summaries of Temporal Scalar Fields

Pressure systems, fire activity, radiation patterns, and pollution concentrations evolve through changes in feature strength, extent, position, and connectivity. A static temporal summary places these changes in one view, allowing events at distant times to be compared directly. Spatial linearization assigns each field to a one-dimensional profile and places time along the orthogonal axis. The central design problem is to compress space while retaining the feature relationships needed to interpret that profile.

1.2 Merge Tree Maps and Reference Displacement

Temporal merge tree maps (TMTM) use augmented merge trees to keep features together, retain scalar context, and optimize temporal overlap [9]. Spatiotemporal merge tree maps (ST-MTM) add a hierarchy-constrained layout that optimizes feature distances and temporal coherence [3]. In both approaches, recursive child-block ordering keeps each subtree contiguous. This constraint encodes scalar connectivity, whereas a geographic reference orders features by their locations in the source domain. The two orders can disagree even when the feature positions and merge tree are known exactly.

Consider two pressure minima that merge within one trough and a third minimum whose projected location lies between them. A contiguous interval for the trough's two minima excludes that reference order. Moving the third minimum outside the interval or moving a trough minimum past it introduces positional error. Placing all three in reference order instead changes the scalar barriers required by their original merge relations. This conflict arises from the hierarchy itself and persists after layout optimization.

1.3 Research Question

We ask how much reference displacement is unavoidable when the merge hierarchy is preserved, and how much merge-level distortion is required to improve that alignment. Answering both parts requires an optimum over admissible layouts and a scalar cost for changed leaf orders. The first distinguishes a structural obstruction from an avoidable layout error; the second gives a common scale for comparing alternative relaxations.

1.4 Contributions

Reference-Guided Merge Tree Maps (RG-MTM) makes a fixed spatial reference the layout target and provides two operating modes. RG-MTM-P preserves the original labelled merge levels. RG-MTM-R relaxes selected subtree-contiguity constraints, retains the extrema and their values, and measures the altered merge levels in the rendered profile.

The paper contributes an attained positional certificate that decomposes interval-packing and hierarchy costs, with feature triples that explain individual conflicts; an optimal fixed-order merge-level construction that defines small-tree position–topology frontiers; and a reference-guided temporal-map pipeline evaluated on six sequences and 342 frames. The experiments jointly assess reference order, achieved anchor displacement, merge-level distortion, geometric motion, and temporal stability, providing evidence for choosing between the two modes.

2 RELATED WORK

2.1 Static Temporal Representations and Spatial Linearization

Hovmöller diagrams summarize variation along a selected spatial direction [8]. Space-filling curves arrange spatial samples along a traversal and can adapt that traversal to field values [6, 18]. Their sample coverage supplies scalar context, while repeated entry into a feature can divide it into several displayed runs. MotionRugs organize trajectories in a compact temporal representation [2], and nested tracking graphs express evolving feature relationships [12]. These representations assign different meanings to their spatial axis: a source coordinate, a traversal position, or a feature order. RG-MTM uses an explicit source-space projection as the reference against which displacement is measured.

2.2 Merge Tree Maps

TMTM traverses augmented merge trees to linearize scalar samples while retaining the merge tree of the field, then aligns consecutive trees by optimizing subtree overlap [9]. ST-MTM separates a distance-aware one-dimensional skeleton from scalar filling: feature intervals, internal tree paths, and borders populate the final profile [3]. We build on this skeleton-and-filling structure. RG-MTM introduces a certificate of reference displacement and separates the tree governing allowable orders from the original tree supplying

scalar values. This separation permits controlled changes in ordering while keeping the original merge levels available as the comparison target.

2.3 Hierarchy-Constrained Ordering and Stability

Optimal leaf ordering selects among permutations consistent with a hierarchy [1]. Interval-separation optimization minimizes displacement subject to non-overlap [11]. RG-MTM combines these constraints: hierarchy determines which orders are admissible, and interval geometry determines the displacement achievable within each order. Its temporal objective follows projected reference motion, so the target can move with the feature. Persistence simplification changes the retained feature set [5]; our relaxation preserves the leaf set and removes selected contiguity constraints. The scalar quantity governing eligibility is the gap between adjacent merge nodes.

2.4 Merge Tree Comparison and Visualization Quality

Labelled merge-tree comparisons quantify disagreement between corresponding merge levels [13, 17]. We use a fixed-label cophenetic discrepancy and derive the minimum possible error for a specified leaf order with fixed extremum values. Projection measures such as stress and trustworthiness assess distances and neighbourhoods in a chosen embedding [14, 16]. ST-MTM also evaluates temporal displacement [3]. These measures complement reference-order and merge-level errors: they evaluate different aspects of spatial compression and temporal layout. Our evaluation retains their separate definitions and reports their outcomes alongside the position–topology trade-off.

3 Background and Problem Formulation

3.1 Augmented Merge Trees and Temporal Features

The input consists of processed scalar fields $F_t : \Omega \to \mathbb{R}$, augmented merge trees T_t, leaf supports, and cross-frame correspondences. For a join tree, leaves represent minima and scalar values increase toward the root as sublevel-set components merge. A split tree is treated by negating the field. Augmentation retains regular samples along tree arcs, providing the scalar values used to fill a profile.

Leaf i at time t has extremum location R_i^t, extremum value $f_t(i)$, support area A_i^t, and support centroid G_i^t. The extremum location defines the reference anchor; the centroid enters the distance objective. Cross-frame matches maximize support-intersection counts using a Hungarian assignment, accept positive overlaps, and give unmatched leaves new track identifiers. All compared methods share these extracted features and matches.

3.2 One-Dimensional Maps and Hierarchy Preservation

A one-dimensional scalar profile ψ_t assigns retained extrema to anchors and fills the intervals between them. Contracting unary chains gives the branching tree used for ordering; regular samples remain available for rendering. A legal permutation $\pi \in \Pi(T)$ places the leaves of every subtree in one contiguous block and can be generated by recursively permuting child blocks.

For distinct critical levels, the subtree-contiguity constraint characterizes orders that admit the original labelled merge levels. Given anchors for leaves i, j, let I_{ij} be the interval between them. Their merge level in a join profile is its highest value on that interval. We measure the maximum discrepancy from the source tree by

$$d_{\mathrm{top}}(\psi, T) = \max_{i<j} \left| \max_{x \in I_{ij}} \psi(x) - f(\mathrm{lca}_T(i,j)) \right|. \tag{1}$$

The one-leaf error is zero. Labels fix the compared extrema, so the measure directly expresses changes in their merge thresholds.

3.3 Spatial Reference and Interval Constraints

A fixed reference map φ assigns $q_i^t = \varphi(R_i^t)$ in a shared canvas $[a, a+L]$. The experiments use a sequence-wide projection direction. Omitting time indices for one frame, feature i has interval centre z_i, width $w_i = cA_i$, and anchor u_i with slack $h_i = \rho w_i/2$. Consecutive leaves i, j in the chosen order satisfy

$$\begin{aligned} z_j - z_i &\geq (w_i + w_j)/2 + g, \\ a + w_i/2 \leq z_i &\leq a + L - w_i/2, \\ |u_i - z_i| &\leq h_i. \end{aligned} \tag{2}$$

The scale c, minimum gap $g > 0$, and $0 \leq \rho \leq 1$ are shared across frames. The packing condition $\sum_i w_i + (n-1)g \leq L$ is checked before optimization. Positive gaps keep anchors ordered even when anchors move within their feature intervals.

3.4 Design Objectives

Reference consistency minimizes the largest anchor displacement $\max_i |u_i - q_i|$. Merge-level fidelity minimizes Equation (1) while retaining extremum values. Temporal continuity encourages matched features to follow their projected source displacement. These objectives use two error domains: position in reference-coordinate units and merge distortion in scalar-field units. The analysis below identifies their attainable optima; the construction then selects coordinates within a positional budget and reports the achieved errors.

4 Position–Topology Analysis

4.1 Positional Certificates and Cost Decomposition

For a fixed leaf order π, define $\tau(\pi)$ as the minimum of $\max_i |u_i - q_i|$ under Equation (2). Eliminating the anchors gives centre windows $|z_i - q_i| \leq \tau + h_i$. Renumber leaves by display rank and let $s_k = (w_k + w_{k+1})/2 + g$ and $S_{ij} = \sum_{k=i}^{j-1} s_k$. The space needed to the left and right of centre k is

$$\begin{aligned} d_k^- &= \textstyle\sum_{r<k} w_r + w_k/2 + (k-1)g, \\ d_k^+ &= \textstyle\sum_{r>k} w_r + w_k/2 + (n-k)g. \end{aligned}$$

When total packing is feasible, the exact optimum is

$$\begin{aligned} \tau(\pi) = \max \Big\{ 0,\ & \max_{i<j} \frac{q_i - q_j + S_{ij} - h_i - h_j}{2}, \\ & \max_k (a + d_k^- - q_k - h_k), \\ & \max_k (q_k - h_k - a - L + d_k^+) \Big\}. \end{aligned} \tag{3}$$

The pair term accounts for the reference separation and required interval spacing. The boundary terms reserve enough room for all preceding and following features. Necessity follows by combining the corresponding windows and separation constraints. For sufficiency, place each interval at its earliest feasible centre: delaying it can only make later upper-window constraints harder to satisfy. The resulting placement attains Equation (3); Supplement S1 provides the complete derivation and the one-leaf case.

Minimizing over legal tree orders and over all permutations gives

$$\begin{aligned} \tau^*(T) &= \min_{\pi \in \Pi(T)} \tau(\pi), \qquad \tau_{\mathrm{free}} = \min_{\pi \in S_n} \tau(\pi), \\ H &= \tau^*(T) - \tau_{\mathrm{free}} \geq 0. \end{aligned} \tag{4}$$

Here S_n contains all leaf permutations. The hierarchy-free optimum τ_{free} measures packing and boundary cost. The difference H measures the additional displacement forced by hierarchy under exactly the same reference and interval geometry. Every tree-preserving layout incurs at least τ^*, and some legal layout attains it. Flattening

a non-root branching node removes a contiguity constraint, enlarges the admissible order set, and cannot increase this optimum.

For the evaluated small trees, enumeration of legal orders computes the continuous certificate. A display-grid solver instead propagates earliest-completion arrays and bisects the feasible error. For binary children A, B, the parent recurrence is $E_S = \min(E_B \circ E_A, E_A \circ E_B)$: each composition places one complete child block after the other. A feasibility test costs $O(nN)$ for n leaves and N pixels; a degree-m node uses $O(2^m mN)$ subset composition. This solver certifies the grid model. Its half-pixel comparison to the continuous model assumes grid-aligned widths and gaps; Supplement S1 gives the rounding bound for real-valued geometry.

4.2 Conflict Triples and Their Interpretation

A conflict triple consists of two leaves i, j belonging to one subtree and an outside leaf k with $q_i < q_k < q_j$. Subtree contiguity forces k before both leaves or after both. The first choice reverses the reference order of i, k; the second reverses that of k, j. Even with zero-width anchors, the maximum displacement is at least $\frac{1}{2}\min(q_k - q_i, q_j - q_k)$. Interval widths, gaps, and canvas boundaries can increase the required displacement.

The triple gives a local explanation for a global certificate: it names the reference relation excluded by a specific subtree. Several overlapping triples can contribute to a stronger obstruction, so the complete optimum is computed over the tree and geometry. In the pressure example, the two trough minima form the subtree pair and the geographically intermediate minimum is the outside leaf. Their scalar merge gap then quantifies the cost of breaking that relation.

4.3 Minimum Merge-Level Distortion

Let v be a non-root branching subtree that becomes non-contiguous, and let $\gamma_v = f(p(v)) - f(v)$ be its parent merge gap. Inserting an outside leaf between two subtree leaves makes their displayed enclosing interval contain a pair whose original merge level is higher. A line profile takes its merge level from the interval maximum, forcing at least one discrepancy of $\gamma_v/2$. Fixed extremum values can require an even larger correction.

The exact fixed-order cost incorporates those endpoint values. Number leaves by their displayed order, set $M_{ij} = f(\mathrm{lca}_T(i, j))$, and define

$$C_k = \min_{i \le k < j} M_{ij}, \qquad \ell_k = \max(f(k), f(k+1)).$$

The lowest pairwise target crossing gap k is C_k, while its barrier must be at least ℓ_k. The optimal error is

$$\delta^*(\pi) = \max\left\{0,\ \max_k(\ell_k - C_k),\ \frac{1}{2}\max_{i<j}\left(M_{ij} - \max_{i \le k < j} C_k\right)\right\}. \tag{5}$$

To see why, an error bound δ requires every gap barrier b_k to obey $\ell_k \le b_k \le C_k + \delta$. Choosing all barriers at their upper limits maximizes every pair's displayed merge level while respecting all upper error bounds. The remaining lower bounds require $M_{ij} - \delta \le \max_{i \le k < j}(C_k + \delta)$, yielding the second term in Equation (5). Thus $b_k = C_k + \delta^*(\pi)$ attains the optimum with fixed extremum values. The result is a minimum for non-strict minima; positive clearance approaches the same infimum when strict minima are required.

4.4 Exact Attainable Frontiers for Small Trees

For a fixed order, interval geometry determines anchor coordinates and scalar barriers determine pairwise merge levels. The two optima can be realized together. Consequently, the closure of attainable position–topology pairs has lower envelope

$$F_T(\delta) = \min_{\pi:\, \delta^*(\pi) \le \delta} \tau(\pi), \qquad F_T(0) = \tau^*, \quad F_T(\infty) = \tau_{\mathrm{free}}. \tag{6}$$

Increasing the scalar budget admits additional permutations, producing a non-increasing frontier. The zero-distortion endpoint enforces the original hierarchy; the unrestricted endpoint retains only the packing constraints.

The parent-gap bound also gives a necessary lower frontier. Flattening all nodes with $\gamma_v \le 2\delta$ yields an order set containing every order achievable within distortion δ. Its positional optimum $\Phi_T(\delta)$ therefore satisfies $\Phi_T(\delta) \le F_T(\delta)$. Fixed leaf values and interactions among broken relations can leave this bound below the exact frontier. Exact evaluation enumerates $n!$ orders and is used as a small-tree validation tool; RG-MTM-R uses selective flattening to choose a practical subset of these orders.

5 Reference-Guided Merge Tree Maps

5.1 Pipeline and Input–Output Structure

RG-MTM takes augmented merge trees, extrema, support regions, and temporal matches. It computes a shared spatial reference, diagnoses each frame, selects an order and anchor layout, fills the complete one-dimensional domain, and stacks the profiles chronologically. The outputs are scalar profiles ψ_t, feature tracks, positional certificates, conflict triples, and achieved position and merge-level errors.

Three objects have distinct roles. The original tree T_t defines the scientific merge targets and supplies augmented samples. The ordering tree T_t' defines admissible leaf permutations and may have fewer contiguity constraints. The filled profile ψ_t is the displayed result whose barriers are measured against T_t. Both operating modes share the reference, geometry, objective, and rendering protocol.

5.2 Fixed Reference and Framewise Diagnosis

For matched extrema $(j, i) \in \mathscr{M}_t$, let $\Delta R^t_{ji} = R^t_i - R^{t-1}_j$. The motion scatter

$$S_R = \sum_{t,(j,i)\in\mathscr{M}_t} \Delta R^t_{ji} (\Delta R^t_{ji})^\top$$

provides a leading unit eigenvector d. Its sign is oriented toward nonnegative total projected motion. Set $q^t_i = \langle R^t_i, d\rangle$, $a = \min_{x\in\Omega}\langle x, d\rangle$, and $L = \max_{x\in\Omega}\langle x, d\rangle - a$. These values fix one reference direction and canvas for the complete sequence. A supplied direction handles degenerate motion and supports comparisons along a chosen geographic axis.

For each frame, the method checks packing and computes τ^*_t, $\tau_{\mathrm{free},t}$, H_t, and explanatory triples from the original tree. These quantities diagnose whether the specified hierarchy forces substantial displacement before a layout is selected. The default conflict trigger is $H_t > \theta$ with $\theta = 0.02L$.

5.3 RG-MTM-P: Layout with Preserved Merge Levels

RG-MTM-P sets $T_t' = T_t$ and uses a positional budget $B_t = \tau^*(T_t) + \beta$. The allowance β gives the secondary objectives room to improve geometry and motion consistency beyond a pure minimax placement. Candidate orders satisfy $\pi \in \Pi(T_t)$ and $\tau(\pi) \le B_t$. For each order, a constrained quadratic program chooses interval centres and anchors under Equation (2) and $|u^t_i - q^t_i| \le B_t$.

The objective is $4E_{\mathrm{ref}} + E_{\mathrm{dist}} + 0.5E_{\mathrm{motion}} + 0.1E_{\mathrm{ecc}}$. Its components average squared reference residuals $u^t_i - q^t_i$, squared differences between ordered anchor distances and source-centroid distances, squared matched motion residuals $(u^t_i - u^{t-1}_j) - (q^t_i - q^{t-1}_j)$, and squared anchor eccentricities $z^t_i - u^t_i$. All terms use common length units; pair and match terms divide by their respective counts.

The lowest objective selects the order and coordinates. Frames are optimized sequentially, using the preceding layout for the temporal term. Original-tree filling then realizes the retained labelled merge levels.

5.4 RG-MTM-R: Relaxation and Budgeted Layout

RG-MTM-R begins with the same diagnosis. If $H_t \leq \theta$, it retains the original ordering tree. Otherwise, it marks non-root branching nodes with $\gamma_v \leq \kappa$ as eligible and initially flattens them all. Flattening removes only their ordering constraints; their original merge values remain in T_t.

The method computes a rounded positional certificate b_0 for this fully eligible relaxation. It then visits eligible nodes in decreasing merge-gap order and restores a constraint whenever the rounded certificate stays within half a grid pixel of b_0. The reference score remains fixed throughout restoration, preventing repeated tolerance from accumulating. This procedure recovers constraints that contribute little positional benefit while retaining the additional orders needed by the conflict.

For the resulting T_t', set $B_t = \tau^*(T_t') + \beta$ and solve the same order-selection and coordinate problem as RG-MTM-P. The evaluated settings are $c|\Omega| = L/2$, $\rho = 0.5$, $g = L/240$, $\beta = L/120$, and $\kappa = 0.2\Delta_F$, where Δ_F is the sequence-wide processed scalar range. Restoration uses a 4096-pixel certificate grid; the final small-tree budget uses continuous enumeration. The parameter κ controls individual merge-gap eligibility. The selected order's total scalar cost is subsequently computed by Equation (5) and measured in the filled profile.

5.5 Scalar Filling and Temporal Assembly

Filling follows the augmented-sample structure of ST-MTM [3]. Place each extremum value at its anchor and distribute resampled leaf-arc values alternately to either side. Fill the remaining internal intervals along paths to the original least common ancestor. Sibling leaves place their merge sample centrally. A direct–deep pair places it next to the direct child. A deep–deep pair allocates space to both paths in proportion to their sample counts, placing the merge sample between them. Border intervals sample the corresponding root paths. These operations assign scalar values across the entire display domain.

In RG-MTM-P, the original LCA barriers reproduce labelled pairwise merge levels. In RG-MTM-R, compute the optimal barriers $b_k = C_k + \delta^*(\pi)$ against T_t. Clip each inter-anchor segment at its barrier and raise its peak when needed, retaining the endpoint extrema and their values. Positive clearance keeps strict extrema where raster capacity permits. This correction minimizes the largest merge-level discrepancy for the selected order.

The renderer stacks filled columns in chronological order on a fixed raster and applies one scalar colour map. It records continuous anchor errors before rounding and actual pairwise merge errors after filling. Nearest-pixel rounding changes an anchor by at most $L/[2(N-1)]$ for N pixels; the renderer checks that distinct anchors and required gaps remain representable. Supplement S2 gives the objective, sample allocation, and raster details.

5.6 Error Annotation and Mode Selection

Each frame is associated with its packing cost, hierarchy cost, conflict identities, achieved anchor displacement, and measured merge-level distortion. Positional annotations use reference units; scalar annotations use the field's physical units and its sequence-normalized range. A certificate identifies the best achievable displacement under the specified constraints, while the measured error identifies the selected layout's result.

RG-MTM-P supports readings that require the retained merge thresholds, such as the level connecting two pressure minima. RG-MTM-R supports position-oriented inspection when altered merge levels are acceptable. Comparing the modes at the same reference makes the exchange explicit: the change in anchor alignment is paired with the actual scalar correction required to obtain it. The ERA5 and OLR cases in Section 7 illustrate two different outcomes of that choice.

6 Experimental Evaluation

6.1 Research Questions

The evaluation examines four questions: how frequently hierarchy forces displacement beyond packing; how much RG-MTM-P and RG-MTM-R improve reference alignment; what merge-level cost accompanies relaxation; and how these changes affect geometric motion, spatial relationships, and temporal stability. Correctness checks connect the theoretical optima to implementation outputs. Parameter controls and timing records establish the tested operating range.

6.2 Data and Shared Inputs

The suite contains four real sequences and two synthetic controls, totalling 342 frames. ERA5 supplies 118 mean sea-level-pressure fields over the North Atlantic and Europe, sampled every 12 hours from 1999-11-17 to 2000-01-14 [7]. Fire contains 30 two-day aggregates of Australian fire radiative power from November–December 2019, transformed as $\log(1 + \mathrm{FRP})$ [6]. OLR contains 60 daily NOAA outgoing-longwave-radiation fields over the tropical Indo-Pacific from 2024-06-01 to 2024-07-30 [10]. PM2.5 contains 30 concentration fields over central China from 2000-01-01 to 2000-01-30 [4].

Ring supplies 40 expanding-Gaussian-ring frames on a 14×14 grid, using the archived TMTM author generator [9]. Gaussians supplies 64 frames on a 64×64 grid, with two fixed centres, one moving centre, and a connecting ridge controlling their merge hierarchy. The real datasets respectively represent low-pressure connectivity, fire hotspots, low-radiation structures, and pollution maxima; the controls isolate expansion and prescribed feature motion.

ERA5, OLR, and PM2.5 use 49×49 interior grids; Fire uses 64×64. The Gaussian smoothing scales for ERA5, Fire, OLR, and PM2.5 are 250, 50.7, 500, and 100 km, respectively. OLR and PM2.5 use the smallest common candidate scale retaining at most eight extrema, and Fire uses a twelve-extremum limit. Actual leaf counts range from 1–8 on ERA5, 6–11 on Fire, 2–8 on OLR, 1–5 on PM2.5, 1–7 on Ring, and 3–4 on Gaussians. Pressure and OLR use join trees; the other fields use split trees. All methods share the processed fields, extracted trees, supports, and temporal correspondences. Supplement S2 specifies source acquisition, cropping, boundary processing, scale selection, and synthetic parameters.

6.3 Comparison Methods

The primary comparison uses TMTM, ST-MTM, RG-MTM-P, and RG-MTM-R with $\kappa/\Delta_F = 0.20$. The baseline implementations follow the papers' described protocols. ST-MTM uses temporal mode, uniform distance weights, $r = 0.95$, and temporal weight 0.5; Ring uses its preset weight 1.5. The two RG-MTM modes share interval geometry and objective weights, making their difference attributable to the relaxation and associated scalar correction.

Supplemental controls include native Hilbert and Morton traversals, packed reference sorting, fixed directions, and exact projection. Native curves retain processed grid samples along their traversal. Packed sorting removes hierarchy while sharing RG-MTM's interval constraints and optimal barrier construction. Exact projection places anchors at the reference coordinates and supplies a positional control without a filled scalar profile. These configurations expose the separate effects of ordering, packing, and scalar filling.

6.4 Evaluation Metrics

Reference-order error and achieved displacement. Normalized Reference-Order Error (NROE), denoted P, measures the strongest reversed reference separation in the displayed anchor order π_t:

$$P_t = \frac{\max\{0, \max_{r<s}(q^t_{\pi_r} - q^t_{\pi_s})\}}{2L}, \qquad P = \frac{1}{K}\sum_t P_t. \tag{7}$$

This is the width-free minimum maximum displacement after monotone calibration, allowing methods with different native scales to be compared by order. A zero score means that reference order is compatible with the display. Stable feature-order tie-breaking permits separate calibrated coordinates for coincident raster anchors. To measure actual RG-MTM coordinates, we also report $E_{\text{pool}} = \sum_{t,i}|u^t_i - q^t_i|/(L\sum_t n_t)$ and $E_\infty = \max_{t,i}|u^t_i - q^t_i|/L$. These are, respectively, the pooled feature-frame mean and the largest continuous anchor error; the certificate τ^* instead minimizes each frame's maximum.

Merge-level error. Normalized Merge-Level Error (NMLE) divides Equation (1) by the fixed sequence-wide processed scalar range Δ_F: $D_t = d_{\text{top}}(\psi_t, T_t)/\Delta_F$. We report $\text{NMLE}_{\text{mean}} = K^{-1}\sum_t D_t$ and $\text{NMLE}_{\max} = \max_t D_t$. Each frame's value is already a maximum over labelled pairs, so the two sequence summaries expose both typical and worst-frame distortion. All reported method errors use measured rendered barriers. The fixed-order optimum provides an independent prediction for the relaxed filling.

Geometric motion proxy. A tracked pair supplies a clear event if its two-dimensional extremum distance changes by more than 2% of the domain diagonal over two steps. The displayed pair-distance change predicts approach or separation, using 2% of the method's sequence-wide occupied anchor range as the no-change threshold. Motion error ME is the fraction of reversed or missed clear events; N_{clear} gives the shared denominator. A sequence with no clear events has an undefined ME. The primary ME values use retained anchor records; Supplement S3 reports a separate raster cross-check. Four-step circular temporal blocks provide the bootstrap interval for paired differences, accounting for reuse of frames in overlapping windows. Because the display threshold depends on occupied range, ME evaluates the complete layout under this protocol.

Geometry and temporal stability. The saved diagnostics comprise order flips, mean matched-anchor displacement normalized by occupied range (J), optimally scaled stress-1 between anchor and extremum distances (S), and nearest-neighbour identity preservation (NN). A separate re-evaluation computes ST-MTM-compatible scale-normalized stress (SNS) from support-centroid distances, trustworthiness (TW) with $k = 1$, and total temporal displacement (TD) after mapping each complete canvas once to $[0,1]$ [3, 15, 16]. SNS averages valid frames with at least two features; TW requires $k < n/2$. TD sums matched displacements in canvas-width units. These standard measures and the saved S/NN/J diagnostics retain their own source locations, scales, and aggregation conventions. Supplement S3 also reports reference-rank correlations and three-class motion classification.

6.5 Correctness Validation

Hierarchy cost exceeds 2% of reference extent in 40/118 ERA5 frames, 19/30 Fire frames, 7/60 OLR frames, 5/30 PM2.5 frames, 21/40 Ring frames, and 47/64 Gaussian frames. Mean hierarchy costs are 4.17%, 6.12%, 0.94%, 1.31%, 2.83%, and 5.95%, respectively. Corresponding mean packing costs are 0.16%, 0.06%, 0.02%, 0.09%, 0.05%, and 0.03%. Thus hierarchy accounts for most certified displacement in each sequence. ERA5's maximum hierarchy cost reaches 25.35%, while its maximum packing cost is 4.70%; these are separate framewise maxima.

Independent optimization and order-enumeration checks validate the positional formulas, grid bounds, and optimal barriers. The retained implementation-validation run passed 25 tests. Across the six sequences, filled-raster merge errors agree with Equation (5) within 1.2×10^{-5} in processed scalar units. TMTM, ST-MTM, and RG-MTM-P have zero measured labelled pairwise merge error on every evaluated frame. For RG-MTM-R, the positional optimum of its selected order lies on the exact position–topology frontier in 339/342 frames. This comparison evaluates the selected framewise order against the exact attainable set; achieved coordinates additionally reflect the budget allowance and secondary objective.

6.6 Position Gains and Merge Costs

RG-MTM-P lowers NROE relative to both baselines on every sequence while preserving labelled merge levels. RG-MTM-R further lowers NROE and achieved pooled displacement on all six sequences. The following results use percentages for NROE, anchor displacement, and NMLE, and decimal fractions for ME.

ERA5. NROE is 19.23% for TMTM, 16.28% for ST-MTM, 3.77% for RG-MTM-P, and 1.16% for RG-MTM-R. Relaxation reduces pooled anchor error from 4.13% to 2.14% and maximum anchor error from 26.19% to 18.46%. Its NMLE is 1.47% on average and 26.07% at the worst frame. ME decreases from 0.471 and 0.413 for the baselines to 0.231 and 0.183 for the two modes, over 104 clear events. Relative to ST-MTM, RG-MTM-R changes 43 errors to 19, a paired difference of -23.1 percentage points with a 95% temporal-block interval of $[-33.6, -12.8]$.

Fire. NROE decreases from TMTM's 29.30% and ST-MTM's 24.02% to 5.83% in RG-MTM-P and 0.34% in RG-MTM-R. Pooled anchor error falls from 3.18% to 0.80%. Relaxation incurs 3.96% mean and 19.56% maximum NMLE. ME is 0.393, 0.357, 0.321, and 0.312 in the same method order, over 112 events. The large alignment improvement from preserving to relaxed mode changes only one motion decision, from 36 errors to 35.

OLR. NROE is 19.31%, 17.21%, 0.89%, and 0.00% for TMTM, ST-MTM, RG-MTM-P, and RG-MTM-R. Relaxation reduces pooled anchor error from 1.00% to 0.57% and maximum anchor error from 17.14% to 1.69%, at 1.10% mean and 33.56% maximum NMLE. ME improves from the baselines' 0.446 and 0.214 to 0.054 in both modes, corresponding to 3/56 errors. This sequence separates improved reference alignment from an unchanged motion-proxy outcome.

PM2.5. NROE is 11.86% for TMTM, 11.30% for ST-MTM, 1.11% for RG-MTM-P, and 0.16% for RG-MTM-R. Pooled anchor error decreases from 2.30% to 1.18%; relaxed NMLE is 0.86% on average and 9.48% at maximum. ME changes from 0.714 and 0.429 for the baselines to 0.429 and 0.286 for the two modes. Only seven clear events are available, so the difference between RG-MTM-P and RG-MTM-R is one decision.

Ring. NROE is 7.18%, 10.39%, 2.66%, and 0.13% in the four-method order. Relaxation reduces pooled anchor error from 2.43% to 0.71% and incurs 2.73% mean and 12.35% maximum NMLE. ME decreases from 0.645 for TMTM and 0.347 for ST-MTM to 0.315 for RG-MTM-P and 0.129 for RG-MTM-R over 124 clear events. The expanding field therefore shows alignment and motion-proxy gains together, with a measurable scalar cost.

Gaussians. Both baselines have 25.76% NROE, compared with 5.43% for RG-MTM-P and 1.38% for RG-MTM-R. Pooled anchor error falls from 4.18% to 1.65%; relaxed NMLE is 3.35% on average and 9.98% at maximum. No pair passes the default clear-event threshold, so this control contributes position and topology evidence without a motion-error rate.

The standard geometry and stability measurements reveal different trade-offs. On Fire, RG-MTM-R improves SNS from ST-MTM's 0.0786 to 0.0579 and TD from 7.9585 to 3.2487, while TW decreases from 0.9195 to 0.8725. On Gaussians, SNS improves from 0.0845 to 0.0638 and TW from 0.7533 to 0.8639, while TD rises from 0.2287 to 0.5396. On ERA5, ST-MTM retains lower SNS (0.0297 versus 0.1097), higher TW (0.9653 versus 0.7554), and lower TD (7.1607 versus 23.8891). It also leads on all three measures for OLR and PM2.5. The position-oriented objective thus improves the designated reference while exchanging other geometric and temporal properties.

6.7 Ablation and Sensitivity

The preserving/relaxed comparison isolates the benefit of additional orders. A filling ablation holds those selected orders fixed and replaces the uncorrected original-LCA filling with optimal barriers. Mean ERA5 NMLE decreases from 2.07% to 1.47%, and Ring decreases from 5.46% to 2.73%. These improvements arise from scalar correction after layout selection.

Packed reference sorting reaches zero NROE on ERA5, with 6.18% mean NMLE and 15/104 motion errors. RG-MTM-R retains a smaller scalar cost of 1.47% mean NMLE while making 19/104 errors. The control therefore identifies a further position-oriented option at a higher merge-level cost. Native space-filling traversals supply an additional comparison based on sample retention rather than RG-MTM's refilling protocol.

Direction controls hold one of eight supplied axes fixed throughout a sequence. Conflict-frame shares vary from 11.0% to 34.7% on ERA5 and from 50.0% to 76.7% on Fire. A separate persistence control removes 34.1% of ERA5 leaves at a 2% scalar-range cutoff, leaving 18.6% of frames conflicting. This changes the feature set, while RG-MTM-R retains its extrema. Further controls vary hierarchy triggers, merge-gap eligibility, extraction, motion thresholds and windows, and ST-MTM parameters. Supplement S3 reports their protocols and full results.

6.8 Computational Performance

The retained binary-tree grid benchmark diagnoses 4096 leaves on 16384 pixels in approximately 3.6 seconds. Its per-feasibility cost is $O(nN)$, with additional bisection iterations for the error budget. At higher-degree nodes, subset composition introduces an exponential dependence on child count.

Complete map construction additionally enumerates legal orders, solves constrained quadratic programs, samples scalar paths, and corrects barriers. The evaluated fields contain at most eleven retained leaves per frame. Exact frontier analysis enumerates all $n!$ permutations and serves as a small-tree reference computation. The available full-pipeline records lack a uniform repeated per-stage timing breakdown; the reported large-tree timing measures diagnosis. Scalable order selection and comparable construction-stage benchmarks are the remaining computational requirements for larger feature populations.

7 Application Cases

7.1 ERA5: Geographic Order and Pressure Connectivity

At ERA5 step 48, corresponding to 1999-12-11 00 UTC, minima near Scotland and Scandinavia merge at approximately 994.1 hPa. A shallower Adriatic minimum lies between them along the fixed spatial reference. The first two minima belong to one low-pressure subtree, making the Adriatic minimum the outside leaf of a conflict triple. The obstruction concerns a specific source relation: the two trough minima should connect before either joins the outside component, but their desired geographic order separates them.

RG-MTM-P keeps the trough's merge threshold and moves at least one anchor away from its reference. RG-MTM-R admits a closer reference order and its optimal filling changes a pairwise merge level by 12.6 hPa. This value is the selected frame's scalar cost in pressure units; the sequence-level mean NMLE of 1.47% averages all 118 frames. Reading the local cost alongside the anchor residuals explains which geographic gain requires an altered connectivity threshold.

For an inspection centred on relative projected location, the relaxed order brings the displayed extrema closer to their geographic reference. For an inspection centred on the pressure level connecting the Scotland and Scandinavia minima, the preserving mode retains the relevant source value. The paired modes make these two readings available with explicit errors, using the same extrema and temporal matches.

7.2 OLR: Alignment Gains without a Motion-Proxy Gain

OLR provides a contrasting case. RG-MTM-P already reduces NROE to 0.89% and makes 3/56 motion errors. RG-MTM-R reaches reference-compatible order, reduces pooled anchor error from 1.00% to 0.57%, and makes the same three errors. Its 1.10% mean NMLE includes a worst-frame cost of 33.56% of the processed radiation range. The improved alignment therefore offers no additional benefit for the evaluated approach/separation decisions.

The local inspection window is anchored at the first maximum-hierarchy-cost frame, a rule determined from the certificate. Within that window, conflict triples identify the excluded reference relations and the two modes expose the resulting anchor and barrier changes. The 3/56 motion rate and 33.56% maximum NMLE remain sequence summaries, rather than values assigned to that local window. For the tested motion task, RG-MTM-P retains the merge thresholds with the same observed error count; a position-specific task may still motivate the relaxed view.

7.3 Relating Profile Changes to Measured Errors

Each case links three quantities: the hierarchy certificate explains why some displacement is unavoidable, the achieved anchor residual states how far the chosen feature moves, and the rendered barrier discrepancy states which merge threshold changes. Temporal tracks connect the same retained features across frames. Keeping these quantities in their respective units lets a reader distinguish a shifted feature, a changed merge level, and a change in pairwise motion without inferring one from another.

8 Discussion and Limitations

8.1 Choosing between Preserving and Relaxed Modes

The certificate gives a decision basis before construction. When H is small, hierarchy contributes little additional reference displacement, making RG-MTM-P a direct choice for retaining merge thresholds. When H is large, the conflict triples identify which relations obstruct alignment and the relaxed result quantifies the scalar price of admitting other orders. ERA5 demonstrates a position and motion-proxy benefit; OLR demonstrates a position gain with unchanged motion decisions.

A scalar tolerance should be checked against achieved $\mathrm{NMLE}_{\max}$. The eligibility threshold κ bounds each candidate node's parent gap; interacting broken relations and fixed endpoint values determine the total error. This explains how the default 20% eligibility setting can produce a 33.56% worst-frame NMLE. Minimax correction can also change a relation inside a still-contiguous subtree. Protecting selected merge levels therefore requires explicit barrier constraints, and a hard total-error budget requires testing candidate orders against $\delta^*(\pi)$.

8.2 Position, Geometry, and Temporal Stability

A one-dimensional reference expresses one selected spatial coordinate. Its order, absolute coordinate error, two-dimensional distances, and neighbourhood ranks encode different properties. The

experiments make their distinction concrete: RG-MTM improves reference order on every sequence, while ST-MTM retains better SNS, TW, and TD on ERA5, OLR, and PM2.5. NROE measures the order's calibrated positional obstruction; achieved anchor errors measure the actual coordinates under finite widths and packing.

Temporal displacement combines true source motion with layout changes. A lower TD can represent a stable display even when projected source motion is suppressed. RG-MTM instead penalizes the residual from that projected motion. Error bounds for projected approach/separation require achieved anchor errors at the relevant frames; displacement perpendicular to the chosen axis remains outside that guarantee. The geometric motion proxy evaluates two-dimensional pair-distance changes under a declared display threshold and complements these coordinate-level checks.

8.3 Preprocessing, Correspondence, and Offline Reference

Smoothing, cropping, boundary processing, and weak-background handling determine the scalar fields and retained merge trees. OLR and PM2.5 smoothing was selected to limit feature counts, and January PM2.5 was chosen after exploring four months. The reported topology and error rates therefore describe those processed sequences. Persistence controls demonstrate that the conflict can remain after feature reduction, while direction controls show that its frequency depends on the selected spatial question.

Tracks are overlap-based computational correspondences. Their shared use controls differences among methods, but physical event interpretation depends on their validity. The automatic direction also uses all matched displacements in the sequence, making the construction offline. A prospective system would need a supplied axis or a direction fitted to an earlier interval. Such a system should evaluate reference selection and tracking errors in addition to layout errors.

8.4 Scalability and User-Task Validation

The grid certificate scales to thousands of binary-tree leaves, whereas complete order enumeration and quadratic layout are evaluated here on small retained trees. Candidate-order search, dynamic programming for the secondary objective, and distortion-constrained pruning are promising routes to larger maps. Higher node degree remains a separate source of combinatorial cost.

The scalar guarantee concerns labelled merge levels and retained extremum values. Resampling and barrier correction can repeat, omit, or alter regular samples; histogram, thresholded area, and finite root-tail fidelity require additional measures. The current computational proxies assume known feature identities and anchors. Reader identification, annotation readability, and task accuracy require user experiments that compare the two modes under explicit position-reading and merge-threshold tasks.

9 Conclusion and Future Work

RG-MTM quantifies reference displacement forced by merge-tree hierarchy and the merge-level changes required to improve it. An attained positional certificate separates packing from hierarchy cost, conflict triples explain individual obstructions, and optimal fixed-order barriers define exact small-tree trade-off frontiers. The preserving and relaxed modes turn this analysis into temporal scalar maps with measured anchor and merge-level errors.

Across six sequences and 342 frames, reference-guided layout improves reference order relative to TMTM and ST-MTM, and relaxation further improves achieved alignment at a dataset-dependent scalar cost. ERA5 and OLR show why mode selection must account for the intended reading: a positional gain can accompany either improved or unchanged motion-proxy performance. Future work will address scalable full construction, hard merge-error constraints, and user-task validation of position and merge-threshold interpretation.

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